CAPSTONE 69

AUDIO STUDY SERIES

Computer Networks Podcasts from Documents

1. Project Overview

The Audio Study Series is an AI-powered educational project that converts Computer Networks study materials into audio podcasts. It helps students understand networking concepts through audio learning instead of reading lengthy study materials.

2. Objectives

- Convert Computer Networks documents into audio podcasts.
- Make learning easier through clear audio explanations.
- Verify the accuracy of AI-generated information.
- Compare learning performance before and after listening to the podcasts.
- Provide a quiz to evaluate students' understanding.

3. Technologies and AI Tools

Tool	Purpose
NotebookLM	Generate audio overviews from source documents.
ChatGPT	Support research, summarisation and quiz creation.
Gemini	Assist with research and content verification.
Gamma	Create the project presentation.
Google Drive	Store and share audio files and project resources.
GitHub	Host project documentation and submission materials.

4. Topics Covered

1. OSI Model and TCP/IP Models
2. IP Addressing and Subnetting
3. Routing and Switching
4. Network Security

5. Project Workflow

1. Collect Computer Networks study documents from reliable and permitted sources.
2. Review the source materials and verify that they are suitable for the project.
3. Upload the selected documents into an AI audio-generation tool.
4. Generate four audio podcasts, one for each selected topic.
5. Listen to each generated podcast and fact-check its content against the original source.
6. Prepare a learning test and a quiz to assess understanding.
7. Compare the learning results before and after listening to the podcasts.
8. Prepare the final presentation and demonstration.

6. Expected Outcomes

- Four AI-generated audio podcasts based on Computer Networks documents.
- A more accessible way for students to revise networking concepts.
- Audio content checked against the original source documents.
- A before-and-after learning analysis based on collected test results.

7. Audio Accuracy Checklist

Checks whether:

- All important topics are included
- Facts match the original documents
- No important information is missing
- Audio is clear and understandable
- No incorrect information is generated

8. Source Documents - Computer Networks

PDF 1: OSI Model and TCP/IP Model
PDF Link: https://drive.google.com/file/d/18ToYCis2I0HQTJ5CLxVDBZj3oUhzxX9s/view?usp=drivesdk

PDF 2: IP Addressing and Subnetting
PDF Link: https://drive.google.com/file/d/18b9OVzmHYRhbBPbk6DnDywONiAZhnuM-/view?usp=drivesdk

PDF 3: Routing and Switching
PDF Link: https://drive.google.com/file/d/18c6nj_D0HaGTt7qxn7fZMnqdS51P9_9f/view?usp=drivesdk

PDF 4: Network Security
PDF Link: https://drive.google.com/file/d/18f8bvFjDLgWYdp9XXuZeohOtla7ijd6h/view?usp=drivesdk

9. AI-Generated Audio Podcasts

The following audio podcasts were generated using AI tools based on Computer Networks study materials. The links below should be replaced with the actual shareable links of the uploaded audio files.

Podcast 1: OSI Model and TCP/IP Model
Audio link: https://drive.google.com/file/d/16aLtuX1tMlaxCxiq-fyWwTbUTOq8hOHj/view?usp=drivesdk

Podcast 2: IP Addressing and Subnetting
Audio link: https://drive.google.com/file/d/16aLtuX1tMlaxCxiq-fyWwTbUTOq8hOHj/view?usp=drivesdk

Podcast 3: Routing and Switching
Audio link: https://drive.google.com/file/d/16MABxeuZaR1WcETrNVOvfVnIDf18UZjp/view?usp=drivesdk

Podcast 4: Network Security
Audio link: https://drive.google.com/file/d/16EFd7SEjGwM1mgXKEaFJPzAwMcsQ22XH/view?usp=drivesdk

10. Before and After Analysis

Before AI Audio	After AI Audio
Reading lengthy documents	Listening to audio summaries
More reading time	Flexible learning
Manual summarization	AI-based summarization
Text-based learning	Audio-based learning

10.1 Quiz & Answer Key

This 10-question evaluation quiz is designed to assess student understanding of the networking content covered in the audio series as well as the project's methodologies.

Student Evaluation Quiz

1. Which of the following is NOT one of the four main topics covered in the Audio Study Series project?

A) OSI Model and TCP/IP Models
B) IP Addressing and Subnetting
C) Object-Oriented Programming & Inheritance
D) Network Security

2. Which AI tool was primarily used to generate the audio overviews from source documents?

A) Gamma
B) NotebookLM
C) GitHub
D) Google Drive

3. In the OSI model, which layer is responsible for end-to-end communication and flow control (e.g., TCP)?

A) Physical Layer
B) Data Link Layer
C) Transport Layer
D) Application Layer

4. What is the primary purpose of Subnetting in IP addressing?

A) To physically encrypt cables
B) To divide a network into smaller subnetworks
C) To convert IPv4 into domain names
D) To increase router hardware speed

5. What is the key distinction between a Router and a Switch?

A) Routers connect different networks (L3); Switches connect devices within a LAN (L2).
B) Switches handle global internet traffic; Routers handle local Bluetooth only.
C) Routers act only as firewalls; Switches generate IP addresses.
D) There is no functional difference.

6. What is a core objective of the Audio Study Series project?

A) Replace physical universities with VR
B) Convert documents into audio podcasts for flexible learning
C) Design physical NIC cards
D) Build a video streaming platform

7. Which AI tool was used specifically for creating the project presentation deck?

A) ChatGPT
B) Gemini
C) Gamma
D) NotebookLM

8. Which CIA triad principle ensures data is only modified by authorized personnel?

A) Confidentiality
B) Integrity
C) Availability
D) Authenticity

9. According to the Audio Accuracy Checklist, which factor is verified?

A) Audio background music quality
B) Facts matching the original source documents
C) File size remaining under 2MB
D) Presentation slide count being over 20

10. What shift was highlighted in the Before and After Analysis?

A) Reading lengthy documents → Listening to audio summaries
B) Manual coding → Hardware assembly
C) Offline textbooks → In-person lectures
D) Text memory → Permanent retention

Answer Key & Explanations

1. **C) Object-Oriented Programming & Inheritance** — The 4 project topics are OSI/TCP-IP models, IP addressing, routing/switching, and network security.
2. **B) NotebookLM** — Specified as the primary AI engine for audio generation from text inputs.
3. **C) Transport Layer** — Layer 4 handles process-to-process communication, reliability, and flow control.
4. **B) To divide a network into smaller subnetworks** — Subnetting improves efficiency, address management, and security.
5. **A) Routers connect different networks (L3); Switches connect devices within a LAN (L2)** — Basic architectural differentiation.
6. **B) Convert documents into audio podcasts for flexible learning** — Core project mandate.
7. **C) Gamma** — Explicitly listed in the AI tools table for presentation building.
8. **B) Integrity** — Protects data against unauthorized modification or deletion.
9. **B) Facts matching the original source documents** — Explicitly required by the accuracy verification process.
10. **A) Reading lengthy documents → Listening to audio summaries** — Highlighted outcome from Section 10 analysis.

11. Conclusion

This project demonstrates how Artificial Intelligence can support education by transforming traditional study documents into audio learning resources. It provides students with an alternative and engaging way to learn Computer Networks. The usefulness of the podcasts will be evaluated through fact-checking and learning assessments.